```
*| > SU_1_3
* |> u_su1_3

*| Version 1.30 created by Andy 1986.09.16

*| Change log

*| 1.30 1986.09.16 Kosher for second processors, network and ADFS
*|                 except there appears to be a major bug somewhere
*| 1.31 1987.03.11 Two small bug fixes, major mods to file access.

*| Note : not kosher for communicator on these counts :
*| (1) needs to replace PROCasm with an 816-able approach
*|     this is fairly easily solved - also see module in TEMP
*| (2) cannot BGET# or BPUT# from &4xx
*|     this could be soluble if we can rely on resident integers being at
*|     PAGE-some defined offset.
*| NB I cannot see a solution to either problem which is portable between
*|    BBC, Master, 2nd 6502 and Communicator.
*|  the implication of the latter statement is that I was being particularly
*|  dim at the time, since (2) is easily solved.


*| Changes @ 1.3 : Creation of PROCbget and PROCbput necessitate all I/O to
*|                 random access files being done through indirection operators
*|                 since PROCs cannot return values through their parameters

AUTO

*| (C) Pennine Software 1986, 1987

MODE7:A%=HIMEM:HIMEM=A%-&3000

PROCsetup
PROCfiles
PROCfirst

REPEAT:PROCinline:UNTILend

PROCnodes
PROCtidyup

PROCcoords
END

DEFPROCskip(K%):LOCAL C%,J%
*******************************************************************************
*|
*| skip : skip K% characters from input file OR point to <cr> if it occurs
*|        before K% characters have been read
*| Locals : C% used to receive characters from file
*|          J% byte count within the field
*|
*******************************************************************************
J%=0:REPEAT:C%=BGET#I%:J%=J%+1:UNTILC%=13ORJ%=K%
IFC%=13THEN PTR#I%=PTR#I%-1
ENDPROC

DEFPROCread6(K%):LOCAL J%
*******************************************************************************
*|
*| read6 : read a string of to six characters into $K% ( padding to right with
*|         spaces and leaving pointer pointing to <cr> if it occurs before six
*|         characters have been read )
*| Local  : J% used as a byte count within the field
*|
*******************************************************************************
J%=-1:REPEAT:J%=J%+1
K%?J%=BGET#I%:UNTIL(K%?J%)=13 ORJ%=5
IF(K%?J%)=13 THEN PTR#I%=PTR#I%-1
IFJ%<5 FORJ%=J%TO5:K%?J%=32:NEXT
ENDPROC

DEFFNreadno(K%,convert%):LOCAL N$,N,C%,J%
*******************************************************************************
*|
```

```
*| readno : read a number from a K% character field ( if <cr> occurs before K%
*|          characters have been read, leave pointer pointing at <cr> ) such
*|          that a blank or null-length field gives result zero
*|          If convert% has bit 0 set then convert to deca-microns
*|          If convert% has bit 1 set then convert from feet
*|          If convert% has bit 2 set then return as a real else an integer
*|
*| Locals : N$ accumulates characters from the line, then used to be EVALled
*|          N  holds the real result of the conversion
*|          C% used to receive BGETted characters from the file
*|          J% used as a byte count within the field
*|
*******************************************************************************
N$="":C%=BGET#I%:IFC%=13THEN PTR#I%=PTR#I%-1:=0
J%=1:REPEAT:N$=N$+CHR$(C%):C%=BGET#I%:J%=J%+1:UNTILC%=13ORJ%=K%
IFC%<>13N$=N$+CHR$(C%)ELSE PTR#I%=PTR#I%-1
IF N$=STRING$(LENN$," ") THEN =0
N=EVAL(N$):IFconvert%AND1 N=100000*N
IFconvert%AND4 IFINFEET% N=N*.3048
IFconvert%AND2 THEN =N ELSE:=INT(N+0.5)

DEF FNrest:LOCAL G%
*******************************************************************************
*|
*| rest : return the rest of the line up to any control character as a string.
*|        note that if the pointer was pointing at <cr> this is a null string
*|        the pointer points at the start of the next line
*| Local    : G% used to receive characters from the file
*| Features : No checking for excessively long lines, so could trap with
*|             "String too long"
*|            Terminated by any control character - desirable but non-standard
*|
*******************************************************************************
rest$="":G%=BGET#I%:IF G%<32 THEN =""
REPEAT:rest$=rest$+CHR$G%:G%=BGET#I%:UNTILG%<32
=rest$

DEF FNyn:LOCAL C%
*******************************************************************************
*|
*| FNyn : wait for user to press Y or N in either case. Return TRUE if it was Y
*| Local   : C% used as a result register for the INSTRing
*|
*******************************************************************************
REPEAT:C%=INSTR("yYnN",GET$):UNTILC%<>0
=C%<3

DEF PROCsetup
*******************************************************************************
*|
*| setup : define fixed constants, title, keywords, flags, offsets, and create
*|         machine code support.
*| Globals : RELEASE    used to print identity info at various stages
*|           end        remains FALSE until program is to finish
*|           KEYS%      points to buffer holding valid keywords
*|           F%, T%     point to buffers for survey station names ( from & to )
*|           STA%       points to buffer for survey station table
*|           A%         retains value of HIMEM from program start
*|           oldI$      string array to hold names of temporarily closed files
*|           oldP%      integer array to hold their PTR# values
*|           UWATER%    flag controls whether data are for dry or diving survey
*|           INFEET%    flag controls whether data are to be converted
*|           DTHETA     amount to be SUBTRACTED from bearings to get true North
*|           DPHI       amount to be subtracted from clino reading to get truth
*|           D%, L%     accumulative total and plan lengths of survey
*|           rest$, COMMENT$  - used throughout to get strings from file
*|
*******************************************************************************
RELEASE=1.30:end=FALSE
FORJ%=0TO1
 PRINTCHR$141;CHR$131;"PENNINE SOFTWARE CAVE SURVEY ";RELEASE
NEXT:PRINT'
DIM KEYS% 56,F% 7,T% 7,W% 15,a% 15,b% 15
DIM oldI$(4),oldP%(4)
STA%=HIMEM:@%=&90A
FORI%=HIMEM TO A%-4STEP4:!I%=0:NEXT
FOR J%=0TO4:oldI$(J%)="        ":NEXT
```

```
$KEYS%="ORIGIN":$(KEYS%+7)="FEET  ":$(KEYS%+14)="METRIC":$(KEYS%+21)="DIVING":$(KEYS%+28)="NORMAL":$(KEYS%-
F%?6=13:F%?7=0:T%?6=13:T%?7=0
UWATER%=FALSE:INFEET%=FALSE
DTHETA=0.:DPHI=0.:D%=0:L%=0
rest$=STRING$(36," "):COMMENT$=rest$
PROCasm
VDU28,0,23,39,2
ENDPROC

DEFPROCfiles
***************************************************************************
*|
*| files : get filenames from user - check they exist ( where possible )
*|
*| Globals : d$          holds path name to reach SU subdirectories.
*|             note that this is restricted by the filing system in use, and
*|             may take various values, including ( for masters ) fs name.
*|           I$          primary input filename
*|           level%      current nesting level of subsidiary input files
*|           O$          output filename - defaulted same as input.
*|           I%, O%      handles of open input and output files
*|
***************************************************************************
ON ERROR OFF:REM error trapping corrupts stack anyway. need M/C.
REPEAT
INPUT"Path name :"d$:IFd$<>""d$=d$+"."
PRINT"Primary input file ";d$;:INPUT"D."I$
I%=OPENIN(d$+"D."+I$):IF I%=0 PRINT''"Unable to find this file"'"Try again."''
UNTILI%<>0
level%=0
INPUT "Output filename "O$
IFO$=""O$=I$
O%=OPENOUT(d$+"O."+O$)
IFO%=0 P.''"Unable to open ";d$;"O.";O$'"possibly the directory does not exist.":END
VDU12,26
FORJ%=0TO1:PRINTCHR$141;CHR$130;:NEXT
VDU28,0,23,39,2
ENDPROC

DEFPROCfirst:LOCAL G%
***************************************************************************
*|
*| first : read set-up from input file - title, origin coordinates
*| Globals : title$      holds a title for the current survey, printed at times
*|           STA%$0      contains the name of the primary origin station
*|           S%          a running count of the survey station number from zero
*|           a%!...      x,y,z metric coordinates of origin ( internal format )
*|
*| Change @ 1.3 : a%! used instead of A%.B%,C%
*|
***************************************************************************
title$=FNrest
PRINTtitle$'STRING$(LENtitle$,"=")
PROCread6(STA%)
STA%?6=13
STA%?7=0:REM No. of references
S%=0:REM station count
PROCskip(6)
!a%=FNreadno(12,1)
a%!4=FNreadno(12,1)
a%!8=FNreadno(12,1)
COMMENT$=FNrest
PRINTCHR$130"ORIGIN at "CHR$134;$STA%
IFLENCOMMENT$PRINTCHR$134COMMENT$
ENDPROC

DEFPROCasm
***************************************************************************
*|
*| asm : assemble machine code support for station name search
*| Globals : code%       points to DIMensioned space for the machine code
*|
*| NB all internal labels may as well be local.
*| NB portability requirements mean we will have to eliminate the use of fixed
*|       location resident integer variables. This shouldn't be a problem.
*|
***************************************************************************
```

```
"CHANGE":$(KEYS%+42)="FILE   ":$(KEYS%+49)="END    "
```

```
DIM code% 127
FOR J% = 0 TO 2 STEP 2
P%=code%
[OPTJ%
LDA &42C \ K% LSB points to station to search for
STA &82
LDA &42D
STA &83
LDA &438 \ N% LSB points to last station in table
STA &80
LDA &439
CLC
ADC #(HIMEM DIV 256)
STA &81
.nextsta%
LDY #5
.nextbyte%
LDA (&80),Y
CMP (&82),Y
BNE notthisone%
DEY
BPL nextbyte%
LDA &81
SEC
SBC #(HIMEM DIV 256)
STA &42D
LDA &80
STA &42C
RTS
.notthisone%
LDA &80
BNE cantbeEND%
LDA &81
CMP &84
BNE notEND%
LDA #&FF
STA &42F
RTS
.notEND%
DEC &81
LDA &80
.cantbeEND%
SEC
SBC #8
STA &80
JMP nextsta%
]
NEXT
ENDPROC

DEFPROCinline:LOCAL K%
*************************************************************************
*|
*| inline : process a line from the current input file
*| Globals : F%, T%     point to buffers for new station names
*|           KEYS%      buffer for keywords to check against "from" station
*| Local   : K%         index of keyword, if matched. -1 => unmatched
*|
*************************************************************************
IFEOF#I%PROCend:ENDPROC
PROCread6(F%)
K%=-1:FORJ%=0TO49STEP7
IF$F%=$(KEYS%+J%) K%=J%/7
NEXT
IFK%<0 PROCdata:ENDPROC
IFK%=0 PROCorigin:ENDPROC
IFK%<3 PROCunits:ENDPROC
IFK%<5 PROCstyle:ENDPROC
IFK%=5 PROCchange:ENDPROC
IFK%=6 PROCnfile:ENDPROC
IFK%=7 PROCend:ENDPROC

DEFPROCend
*************************************************************************
*|
*| end : process the END keyword in the input file
*| Globals : I%         handle of currently open input file
```

```
*|           level%      current depth of nesting - if zero then this is the
*|                        primary input file that we have open.
*|           end         global flag to stop 1st phase of SU on closing primary
*|           K%          keyword index. will be 7 if we are here because of an
*|                       END keyword, will be some other value if the file just
*|                       ended on us.
*| BUG fix 1987.001      K% might have a value > -1 if the last line contained
*|                       a keyword. Condition changed to K%=7, from K%>0
*|
*****************************************************************************
CLOSE#I%
IF level%=0 end=TRUE:ENDPROC
IF K%=7 PRINT"END occurs in nested file "I$'"Halt processing at this level ?";:IF FNyn end=TRUE:ENDPROC
level%=level%-1
I$=oldI$(level%)
I%=OPENIN(d$+"D."+I$)
PTR#I%=oldP%(level%)
ENDPROC

DEFPROCnfile
*****************************************************************************
*|
*| nfile : save place in old file and open nested file for input
*| Globals : level%     current nesting level. limited arbitrarily to 4
*|           oldI$, I$  old and new filenames
*|           oldP%      PTR# of old file, so we can resume input later
*|
*****************************************************************************
IFlevel%>4 PROCthrowwobbler(1):ENDPROC
oldI$(level%)=I$:I$=FNrest
oldP%(level%)=PTR#I%:CLOSE#I%
I%=OPENIN(d$+"D."+I$)
IFI%=0 PRINT"Unable to open new input file"'d$+"D.";I$:I$=oldI$(level%):I%=OPENIN(d$+"D."+I$):PTR#I%=oldP%(l
level%=level%+1:ENDPROC

DEFPROCorigin
*****************************************************************************
*|
*| origin : read data defining alternate origin
*| Globals : T%          station name for new origin is read into "to" buffer
*|           X%, Y%, Z% x,y,z coordinates of new origin (internal format)
*|           U%          index of new station name in existing list ( should be
*|                        -1, if not, user is redefining an existing station )
*|           S%          internal number for new survey station
*|          STA%$U%      new name added to list
*|          STA%?(U%+7)  count of references to new station - initially zero
*|           W%          32 bit value with high word set to -1 and low word set
*|                       to station no.: S%. Forms from/to for origin definition
*| Note : W% is set to U%/8-65536, not S%-65536, since "origin already defined"
*|          is treated as a soft error, so U%=S%*8 not always true !
*|
*| Bug fix 1987.003      N% assigned to valid value before calling FNscan.
*|                       I cannot see how the program ever worked before.
*|
*| Change @ 1.31 : use of PROCbput. eliminates W% in this context.
*|
*|
*****************************************************************************
PROCskip(30):COMMENT$=FNrest
PROCread6(T%):PROCskip(6)
X%=FNreadno(12,5)
Y%=FNreadno(12,5)
Z%=FNreadno(12,5)
N%=8*S%
U%=FNscan(T%)
IF U%<0 S%=S%+1:U%=S%*8:$(STA%+U%)=$T%:STA%?(U%+7)=0 ELSE PROCthrowwobbler(2)
*|W%=U%/8-65536
PROCbput(O%,4,U%/8-65536,X%,Y%,Z%)
*|FORJ%=&45C TO&46B:BPUT#O%,?J%:NEXT
IFLENCOMMENT$PRINT'CHR$132COMMENT$
COMMENT$=FNrest
ENDPROC

DEFPROCstyle
*****************************************************************************
*|
*| style : process keywords to change type of input data
```

```
):PRINT"Input continues from ";d$;"D.";I$:ENDPROC
```

```
*| Globals : K%        index to keyword - must be 3 for DIVING, 4 for NORMAL
*|           UWATER%   flag to indicate current style
*|
*******************************************************************************
UWATER%=(K%=3)
IFUWATER% PROCdive:PROCskip(12) ELSE PROCskip(30)
COMMENT$=FNrest:IFLENCOMMENT$PRINT'CHR$129COMMENT$
ENDPROC

DEFPROCdive
*******************************************************************************
*|
*| dive : start surveying underwater type of data - get starting info
*| Globals : T%        points to buffer for starting station name
*|           Q%        internal station index for starting station
*|           depth     depth of most recently defined diving survey station
*|
*| Bug fix 1987.003    N% assigned before calling FNscan
*|
*******************************************************************************
PROCread6(T%)
N%=8*S%
Q%=FNscan(T%)
IFQ%<0 PROCthrowwobbler(3)
depth=FNreadno(6,7)
ENDPROC

DEFPROCchange
*******************************************************************************
*|
*| change : accept data to define new instrument calibrations
*| Globals : DTHETA    amount to be subtracted from compass to get true north
*|           DPHI      amount to be subtracted from clino reading to get truth
*|
*******************************************************************************
PROCskip(12)
DTHETA=FNreadno(6,2)
DPHI=FNreadno(6,2)
PROCskip(6)
COMMENT$=FNrest
IFLENCOMMENT$PRINT'CHR$134COMMENT$
ENDPROC

DEFPROCthrowwobbler(E%):LOCAL@%
*******************************************************************************
*|
*| throwwobbler : handle errors
*| Globals : T%        points to buffer holding erroneous station name
*| Locals  : E%        error number
*|           @%        why is this LOCAL ?
*|
*******************************************************************************
PRINT'STRING$(80,"*"):VDU7,7
PRINT" Error on input"'
IF E%=1 PRINT"Input file nesting level too great"'"( maximum depth 4 )"
IF E%=2 PRINT"New origin redefines existing station : ";$T%
IF E%=3 PRINT"Depth change > leg length in dive"
IF E%=4 PRINT"Diving survey start station ";$T%," undefined or missing"
PRINT'STRING$(80,"*"):VDU7
IF E%<4 STOP ELSE ENDPROC

DEFPROCdata
*******************************************************************************
*|
*| data : read and convert a survey leg - of either style
*| Globals : F%, T%    pointers to buffers for from and to station names
*|           G%, U%    pointers to from and to station names in main list
*|          L, THETA, D values of true length, true bearing and plan length.
*|           X%, Y%, Z% x,y,z vector of current survey leg
*|           D%, L%    cumulative plan and total length of survey
*|
*| Change @ 1.3 : Use of PROCbput. eliminates X%, Y%, W% in this context
*|
*******************************************************************************
PROCread6(T%)
PROClist(F%,T%)
IFPOS>34PRINT
```

```
IFU%>G% PRINT;$T%;" "; ELSE PRINT;$F%;" ";
L=FNreadno(6,7)
THETA=FNreadno(6,2)-DTHETA
IF UWATER% PROCdiving ELSE PROCdry
*|X%=INT(D*SINRAD(THETA)+0.5)
*|Y%=INT(D*COSRAD(THETA)+0.5)
D%=D%+INT(D+0.5)
L%=L%+INT(L+0.5)
COMMENT$=FNrest
*|FOR J%=&45C TO &46B
*|BPUT#O%,?J%:NEXT
PROCbput(O%,4,8192*G%+U%/8,INT(D*SINRAD(THETA)+0.5),INT(D*COSRAD(THETA)+0.5),Z%)
IFLENCOMMENT$PRINT'CHR$131COMMENT$
ENDPROC

DEFPROCdry
**************************************************************************
*|
*| dry : handle data for above water surveying style
*| Globals : PHI, PLUMB values of corrected clino reading and vertical plumb
*|           L, D, Z%   true length, plan length and vertical component of leg
*|
**************************************************************************
PHI=FNreadno(6,2)-DPHI
PLUMB=FNreadno(6,7)
D=L*COSRAD(PHI)
Z%=INT(L*SINRAD(PHI)+PLUMB+0.5)
ENDPROC

DEFPROCdiving
**************************************************************************
*|
*| diving : handle data for underwater surveying style
*| Globals : depthF, depthT     depth of from and to stations
*|           Q%, G%, U% pointers to station names for 'last new', 'from' & 'to'
*|           Z, L       depth change and true length of survey leg
*|
**************************************************************************
depthF=FNreadno(6,7)
depthT=FNreadno(6,7)
IFQ%=G%AND depthF=0 depthF=depth ELSEIFQ%=U%AND depthT=0 depthT=depth
IFQ%=G% depth=depthT:Q%=U% ELSE IFQ%=U% depth=depthF:Q%=G% ELSE IF G%>U% depth=depthF:Q%=G% ELSE depth=depthT:Q
Z=depthF-depthT
IFABSZ>L PROCthrowwobbler(3)
D=SQR(L*L-Z*Z)
Z%=INT(Z+0.5)
ENDPROC

DEFPROClist(F%,T%):LOCAL N%:N%=8*S%
**************************************************************************
*|
*| list : check both survey stations for existence in our list. Add them if
*|        they aren't already there, and return internal station numbers
*| Globals : F%, T%     pointers to buffers holding 'from' and 'to' names
*|           G%, U%     offsets to location of these names in the internal list
*|           S%         number of current highest station
*| Local   : N%         offset to next free space in internal station list
*|
*| BUG fix 1987.002     If both stations are undefined, they both got added to
*|          the list in the same place - ie. 'from' station failed to be added.
*|          fixed by incrementing N% if from station has to be added.
*|
**************************************************************************
G%=FNscan(F%)
U%=FNscan(T%)
*|IF G%<0 S%=S%+1:G%=N%+8:$(STA%+G%)=$F%:STA%?(G%+7)=1 ELSE STA%?(G%+7)=STA%?(G%+7)+1
IF G%<0 S%=S%+1:G%=N%+8:N%=G%:$(STA%+G%)=$F%:STA%?(G%+7)=1 ELSE STA%?(G%+7)=STA%?(G%+7)+1
IF U%<0 S%=S%+1:U%=N%+8:$(STA%+U%)=$T%:STA%?(U%+7)=1 ELSE STA%?(U%+7)=STA%?(U%+7)+1
*|W%=8192*G%+U%/8
ENDPROC

DEFFNscan(K%)
**************************************************************************
*|
*| scan : call machine code support to search for a station name
*| Note : uses knowledge of absolute location of resident integers - UGH !
*|
```

```
******************************************************************************
CALL code%
=K%

DEFPROCunits
******************************************************************************
*|
*| units : interpret FEET or METRIC keywords
*| Globals : K%          index to keyword used: 1 => INFEET, 2 => METRIC
*|           INFEET%     flag - true if we are converting from feet
*|
******************************************************************************
INFEET%=(K%=1):PROCskip(30)
COMMENT$=FNrest:IFLENCOMMENT$PRINT'CHR$129COMMENT$
ENDPROC

DEF PROCnodes
******************************************************************************
*|
*| nodes : find all the nodes in the survey
*| Globals : d$, O$      pathname for all files, Output filename.
*|           O%          handle for outpt to node file
*|           J%, I%      count of stations, bytes
*|           J%?7        number of references to this station
*|           R%          offset to start of main station table in the file
*|           S%          station number of final station
*| on exit   R%          total number of nodes found
*|
*| Change @ 1.3 : Use of PROCbput.
*|
******************************************************************************
CLOSE#O%:O%=OPENOUT(d$+"N."+O$)
FORI%=0TO7:BPUT#O%,0:NEXT
FORI%=0TO7:BPUT#O%,STA%?I%:NEXT
FORJ%=STA%+8TOSTA%+8*S%STEP8
IFJ%?7<>2 FOR I%=0TO7:BPUT#O%,J%?I%:NEXT
NEXT
R%=PTR#O%
FORJ%=0TO8*S%+7
BPUT#O%,STA%?J%
NEXT
PTR#O%=0
*|FORJ%=&444 TO&44B:BPUT#O%,?J%:NEXT
PROCbput(O%,2,R%,S%,0,0)
CLOSE#O%
R%=R%/8-1
ENDPROC

DEF PROCtidyup
******************************************************************************
*|
*| tidyup : close files, print statistics and get user to input setup info for
*|          final stage of survey
*| Globals : S%, R%, D%, L% - see caption printed with each one
*|
******************************************************************************
@%=&90A:CLOSE#0:REM just to make sure
VDU12,26
FORJ%=0TO1:PRINTCHR$141;CHR$134;:NEXT
VDU28,0,23,39,2
PRINT'CHR$134;"STATISTICS :"
PRINT'"No. of survey stations ";S%+1
PRINT'"No. of nodes in survey ";R%
@%=&2030A
PRINT'"Total plan length   "D%/100000;" m"
PRINT'"Total survey length "L%/100000;" m"''
SOUND1,-7,170,20
PRINT" DO YOU WANT OUTPUT PAGED ? ";:IF FNyn VDU14 ELSE VDU15
PRINT''" DO YOU WANT OUTPUT PRINTED ? ";:IF FNyn VDU2 ELSE VDU3
CLS
PRINT" ";LEFT$(title$,38)'" ";STRING$(LENLEFT$(title$,38),"=")
IFLENtitle$>38 PRINT" ";MID$(title$,39)'" ";STRING$((LENtitle$)-38,"=")
ENDPROC


DEFPROCcoords
******************************************************************************
```

```
:oords : calculate and display coordinates and closure errors
:lobals : d$, O$      path name for SU files, output filename to use
          O%, S%      handle for C. file, number of stations to reserve space
          STA%        base of internal station table
          A%, B%, C%  x,y,z coordinates of primary origin of survey
          I%          handle for input from previous output ( vector ) file

:hange @ 1.3 : Use of PROCbput

:*************************************************************************
)PENOUT(d$+"C."+O$):PTR#O%=12*S%-1:BPUT#O%,0:PTR#O%=0
:=HIMEM
:?7=1:FORJ%=15TO12287STEP8:STA%?J%=0:NEXT
.I("SPOOL "+d$+"P."+O$):@%=&20309
)RJ%=&404TO&40F:BPUT#O%,?J%:NEXT:L%=0
:bput(O%,3,A%,B%,C%,0)
)PENIN(d$+"O."+O$)
IT''"Station     East/m  North/m Height/m"'"___________________________________"
IT$STA%;TAB(6),A%/100000,B%/100000,C%/100000
:AT:PROCvector:UNTILEOF#I%
:cotidy
;E#I%:CLOSE#O%:*SPOOL
'ROC

'ROCvector
:*************************************************************************

/ector : calculate a new coordinate or closure from vector in file, or warn
        user that his stations are undefined
:lobals : W%          pointer to  16 byte buffer.
           word 0 :   hiword is 'from', lo is 'to' station no.
           word 1-3   x,y,z vector for this survey leg
          F%, T%      internal station numbers derived from W% in vector file
          G%, U%      offsets to station names in table
          G%?7, U%?7 specify whether each station is defined yet

:hange @ 1.3 : Use of PROCbget

:*************************************************************************
)R J%=&45C TO&46B:?J%=BGET#I%:NEXT
:bget(I%,4,W%)
%=W%DIV65536:T%=W%AND65535
(!W%)DIV65536:T%=(!W%)AND65535
STA%+8*F%:U%=STA%+8*T%
%<0 PROCneworg:ENDPROC
G%?7+2*U%?7
X%=1 PROCnewxyz(F%,T%, 1):PROCprint(U%):ENDPROC
X%=2 PROCnewxyz(T%,F%,-1):PROCprint(G%):ENDPROC
X%=3 PROCclosed(F%,T%):ENDPROC
NT$G%;" TO ";$U%;" is undefined."
?ROC

 PROCnewxyz(D%,E%,V%)
**************************************************************************

newxyz : calculate new coordinate
Globals : L%          internal station number of last defined ( retained )

Change @ 1.3 : Use of PROCbput and PROCbget - with indirections

**************************************************************************
F D%<>L% THEN PTR#(O%)=12*D%:FORJ%=&404 TO&40F:?J%=BGET#O%:NEXT
D%<>L% THEN PTR#(O%)=12*D%:PROCbget(O%,3,a%)
%=A%+V%*X%:B%=B%+V%*Y%:C%=C%+V%*Z%
=!a%+V%*W%!4:a%!4=a%!4+V%*W%!8:a%!8=a%!8+V%*W%!12
%=E%:PTR#O%=12*E%:FORJ%=&404 TO&40F:BPUT#O%,?J%:NEXT
E%:PTR#O%=12*E%:PROCbput(O%,3,!a%,a%!4,a%!8,0)
PROC

PROCneworg
**************************************************************************

neworg : define coordinates of alternate origin

**************************************************************************
%=X%:B%=Y%:C%=Z%:L%=T%
```

```
!a%=W%!4:a%!4=W%!8:a%!8=W%!12:L%=T%
*|PTR#O%=12*T%:FORJ%=&404 TO&40F:BPUT#O%,?J%:NEXT
PROCbput(O%,3,!a%,a%!4,a%!8,0)
PRINT"New origin :":PROCprint(U%)
ENDPROC

DEFPROCprint(G%)
*****************************************************************************
*|
*| print : print a set of coordinates
*|
*****************************************************************************
G%?7=1
PRINT$G%;TAB(6),!a%/100000,a%!4/100000,a%!8/100000
ENDPROC

DEFPROCclosed(D%,E%)
*****************************************************************************
*|
*| closed : calculate and print a misclosure
*|
*****************************************************************************
*|IFD%<>L% THEN PTR#(O%)=12*D%:FORJ%=&404 TO&40F:?J%=BGET#O%:NEXT
IFD%<>L% PTR#(O%)=12*D%:PROCbget(O%,3,a%)
*|A%=A%+X%:B%=B%+Y%:C%=C%+Z%
A%=!a%+W%!4:B%=a%!4+W%!8:C%=a%!8+W%!12
*|PTR#O%=12*E%:FORJ%=&460 TO&46B:?J%=BGET#O%:NEXT
PROCbget(O%,3,W%+4)
PRINT$U%;TAB(6),A%/100000,B%/100000,C%/100000
*|A%=A%-X%:B%=B%-Y%:C%=C%-Z%
!a%=A%-W%!4:a%!4=B%-W%!8:a%!8=C%-W%!12
*|PRINT"previous:"X%/100000,Y%/100000,Z%/100000
PRINT"previous:"W%!4/100000,W%!8/100000,W%!12/100000
*|PRINT"MISCLOSES"A%/100000,B%/100000,C%/100000
PRINT"MISCLOSES"!a%/100000,a%!4/100000,a%!8/100000
L%=-1:ENDPROC

DEFPROCcotidy:LOCAL K%
*****************************************************************************
*|
*| cotidy : write out 'undefined' data for any undefined stations
*|
*****************************************************************************
@%=&90A:ENDPROC
*|A%=0:B%=0:C%=&80000000
!a%=0:a%!4=0:a%!8=&80000000
FORJ%=0TOS%
*|IF(8*J%?7)=0 THEN PTR#(O%)=(12*J%):FORK%=&404 TO&40F:BPUT#O%,?J%:NEXT
IF(8*J%?7)=0 THEN PTR#(O%)=(12*J%):PROCbput(O%,3,!a%,a%!4.a%!8,0)
NEXT
ENDPROC

DEFPROCbput(O%,N%,A%,B%,C%,D%):LOCAL J%
!b%=A%:b%!4=B%:b%!8=C%:b%!12=D% :REM formal parms would be OK from BASIC 3
FOR J%=0 TO 3*N%-1
 BPUT#O%,b%?J%
NEXT
ENDPROC

DEFPROCbget(I%,N%,a%)
FORJ%=0 TO 3*N%-1
 a%?J%=BGET#I%
NEXT
ENDPROC
```